	speed needed to leave the planet, the rock will return to the surface of the planet.
3(a)	The internal energy U of an ideal gas is the sum of a random distribution of kinetic ~~and potential~~ energies associated with the molecules or particles of the gas. *Note: Omitting the idea of a "sum" will be incorrect.*
3(b)(i)	Use $pV = nRT$, at constant pressure, $T \propto V$, $\frac{T_2}{T_1} = \frac{V_2}{V_1}$ $T_2 = \frac{V_2}{V_1} \times T_1$ $= \frac{3.6 \times 10^{-3}}{3.2 \times 10^{-3}} \times (12 + 273.15)$ $= 320.79$ K $= 47.63\ ^\circ\text{C}$ $\approx 47.6\ ^\circ\text{C}$
3(b)(ii)	Work done against the atmosphere during expansion $W = p\Delta V$ $= 1.0 \times 10^5 \times (3.6 - 3.2) \times 10^{-3}$ $= 40$ J
3(c)(i)	Use the First Law of Thermodynamics, $\Delta U = Q + W$ $= 101 - 40$ $= 61$ J

No.	**Solution**
3(c)(ii)	Use $pV = NkT$, $N = \dfrac{pV}{kT}$ $= \dfrac{1.0 \times 10^{5} \times 3.2 \times 10^{-3}}{1.38 \times 10^{-23} \times (12 + 273.15)}$ $= 8.132 \times 10^{22}$ molecules Average increase in kinetic energy of a molecule of the gas $= \dfrac{61}{8.132 \times 10^{22}}$ $= 7.501 \times 10^{-22}$ $\approx 7.50 \times 10^{-22}$ J
4(a)(i)	Amplitude, $x_0 = 2.50$ cm